(*) Let A be any set.

$$G = \{ f : A \to A \mid f \text{ is a bijection} \}$$

Prove that $(G, \circ)$ is a group.

Pf: (1) Closure: Let $f, g \in G$

we need, $f \circ g \overset{?}{\in} G$

as f, g are bijective fun,

$$\begin{tikzcd} A \arrow[r, "f"] & A \arrow[dl, "f\circ g"] \\ A \arrow[u, "g"] & \end{tikzcd}$$

$f \circ g$ is also bijective.

We first show that $f \circ g$ is 1-1:

$f \circ g(x) = f \circ g(y)$ for $x, y \in A$

$\Rightarrow f(g(x)) = f(g(y))$, f is bijective $\Rightarrow f$ is 1-1

$\Rightarrow g(x) = g(y)$

$\Rightarrow x = y \Rightarrow f \circ g$ is 1-1

(*) $f \circ g$ is onto:

$f\circ g : A \longrightarrow A$

$\qquad\qquad\qquad\quad \uparrow y.$

f is bijective $\Rightarrow$ f is onto.

$\exists\, x_1 \in A$ s.t. $f(x_1) = y$. —— (I)

let $y \in A$,

From (I)

$f(x_1) = y$

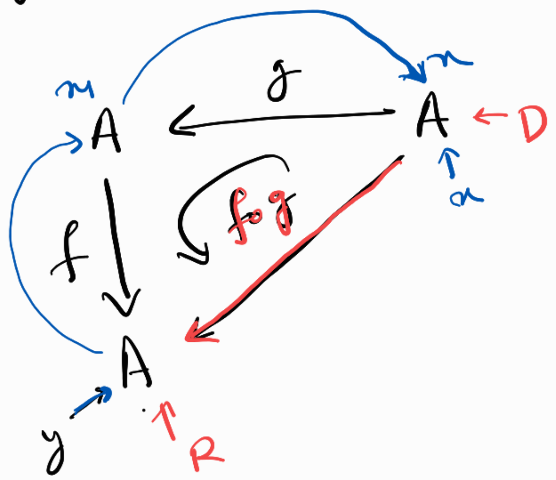
$f : A \longrightarrow A$, $x_1 \mapsto y$

$g : A \longrightarrow A$, $x \mapsto x_1$

$g(x) = x_1$ —— (II)

$\Rightarrow f(g(x)) = y$ [from (II)]

$\Rightarrow f\circ g(x) = y.$



$(f\circ g)^{-1} = g^{-1}\circ f^{-1}$

$\Rightarrow f\circ g \in G$

(II) $f, g, h \in G$, $f\circ(g\circ h) = (f\circ g)\circ h$. (why?)

$\forall\, x \in A$, $f\circ(g\circ h)(x) \equiv f((g\circ h)(x))$

$= f(g(h(x)))$

$(f\circ g)\circ h(x) = (f\circ g)(h(x))$

$= f(g(h(x)))$

$\Rightarrow f\circ(g\circ h) = (f\circ g)\circ h$

(III) $Id_A : A \longrightarrow A$

$x \longmapsto x$

$\boxed{Id_A(x) = x}$, $Id_A \in G$.

we want to show $f \circ Id_A = f$

$\forall x \in A, f \circ Id_A(x) = f(\underbrace{Id_A(x)}_{= x}) = f(x)$

$\Rightarrow f \circ Id_A = f$

$(Id_A \circ f)(x) = Id_A(f(x)) = f(x)$

$\Rightarrow \boxed{Id_A \circ f = f = f \circ Id_A}$

(IV) f bijective map, $\exists f^{-1} : A \to A$.

$f \circ f^{-1}(x) = f(f^{-1}(x)) = x = Id_A(x)$

$\Rightarrow f \circ f^{-1} = Id_A$

$f^{-1} \circ f(x) = f^{-1}(f(x)) = x = Id_A(x)$

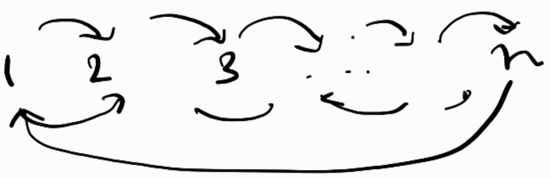
$\Rightarrow \boxed{f^{-1} \circ f = Id_A}$

$\Big(G = \{ f : \underset{\substack{\downarrow \\ \text{any set}}}{A} \to A \mid f \text{ is bijection} \}, \circ \Big)$ is a group.

- particularly, $A = \{1, 2, 3, \dots, n\} = \{a_1, a_2, \dots, a_n\}$

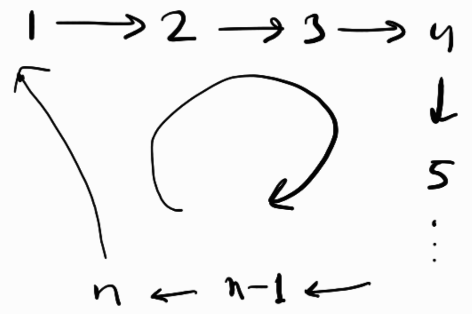
  $\rightsquigarrow G = \{ f : A \to A \mid f \text{ is bijection} \} = S_n$ (Symetric group/permutation group on n element set)

- order $(S_n) = |S_n| = o(S_n) = n!$



$\sigma \in S_n$

$$\sigma(1) = 2$$
$$\sigma(2) = 3$$
$$\vdots$$
$$\sigma(n) = 1$$



$$\sigma = \begin{pmatrix} 1 & 2 & 3 & \cdots & n-1 & n \\ 2 & 3 & 4 & \cdots & n & 1 \end{pmatrix}$$

$$\rho = \begin{pmatrix} 1 & 2 & 3 & 4 & \cdots & n-1 & n \\ 2 & 1 & 4 & 3 & \cdots & n & n-1 \end{pmatrix} \to n \text{ even}$$

$$S_3, \quad \rho = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$

$$P_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} = P_1^{-1}$$

$$P_2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

$$P_2^{-1} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$P_2(1) = 2 \Rightarrow P_2^{-1}(2) = 1$$
$$P_2(2) = 3 \Rightarrow P_2^{-1}(3) = 2$$

$\downarrow P_2(3) = 1 \Rightarrow P_2^{-1}(1) = 3$

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$P_2 = \begin{pmatrix} 1 & 2 & 3 \\ \updownarrow & \updownarrow & \updownarrow \\ 2 & 3 & 1 \end{pmatrix}$$

$$P_2^{-1} = \begin{pmatrix} 2 & 3 & 1 \\ 1 & 2 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$P_3 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

$$P_3^{-1} = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 2 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

$$P_4 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

$P_4^2 =$

$$\begin{tikzcd} 2 \arrow[d, bend left] \\ 3 \arrow[u, bend left] \end{tikzcd}$$

$$P_4^{-1} = \begin{pmatrix} 1 & 3 & 2 \\ 1 & 2 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

$$P_6 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$\begin{array}{c} 1 \rightarrow 3 \\ 3 \\ \swarrow \\ 2 \end{array}$$

$\rightarrow P_6^3 = \mathrm{Id}$!

$$P_6^{-1} = \begin{pmatrix} 3 & 1 & 2 \\ 1 & 2 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$